

VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY
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SEMINAR DOCUMENT

PROGRESS REPORT

Team Members

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Version	Date	Author	Description
1.0	2025-06-10	Cao Uyển Nhi	Initial template creation
1.1	2025-06-12	Cao Uyển Nhi	Added content structure and sections

Note: Please remember to update these sections whenever there are changes to the document content.

Chapter 1

Introduction

1.1 Project Overview

The Performance Testing Tools seminar project is a comprehensive study of performance testing tools, focusing on Apache JMeter and AI applications in performance testing. The project aims to explore the basic concepts of Performance Testing (Load Testing, Stress Testing, Spike Testing, Endurance Testing), compare popular tools in the market, and conduct a practical demo with JMeter on the JPetStore system.

The importance of this project is shown by how Performance Testing plays a key role in ensuring software quality, especially in the digital age when web applications must serve millions of users at the same time. Research and application of AI in Performance Testing is also a new trend, helping to automate result analysis and detect system issues.

1.2 Objectives of the Progress Report

This Phase 1 progress report has the following main objectives:

- **Report implementation progress:** Evaluate the completion status of tasks in Phase 1 (Foundation & Research) during the first 3 weeks
- **Present research results:** Summarize knowledge collected about Performance Testing, tool comparisons, and JMeter research
- **Assess output quality:** Check the quality of completed deliverables such as planning files, research documents, and tool comparison tables
- **Identify problems and risks:** Find difficulties encountered during implementation and propose solutions
- **Plan Phase 2:** Prepare for the next stage (Scenario Design & Testing)

1.3 Scope of the Project

Within project scope:

- Research theory about Performance Testing (Load, Stress, Spike, Endurance Testing)
- Compare Performance Testing tools: JMeter, NeoLoad, WebLOAD, LoadUI, LoadRunner
- Deep research on Apache JMeter (architecture, components, installation)
- AI applications in Performance Testing (anomaly detection, automated analysis)
- Conduct demo on JPetStore system
- Design and execute test scenarios for business flow: Register → Login → Purchase → Payment

Outside project scope:

- No deep research on Security Testing or other testing types
- No development of new automated tools
- No testing on real production systems
- No coverage of performance testing for mobile applications

1.4 Team and Stakeholders

Role	Member	Responsibility
Team Leader	Cao Nhi	Planning, AI research in Performance Testing, analysis evaluation, presentation MC
Member	Thanh Thuý	Basic research on Performance Testing and business flow analysis
Member	Minh Trí	Performance Testing tool comparison and JMeter test script writing
Member	Cát Tường	Deep JMeter research (theory) and test plan design
Member	Việt Tú	JMeter installation setup, test execution, and demo video recording

Stakeholder Type	Stakeholder	Interest/Expectation
Instructor	Course instructor	Evaluate research quality, practical application, presentation skills
Audience	Other student groups in class	Learn new knowledge about Performance Testing, reference research methods
Research Community	Software Testing research community	Scientific value of research, knowledge contribution about AI in Performance Testing

1.5 Timeline Summary

The project is divided into 3 main phases over 9 weeks:

Phase	Timeline	Activities	Status
Phase 1 - Foundation & Research	Week 1-3	Complete theoretical research, tool comparison, JMeter environment setup	Done
Phase 2 - Scenario Design & Testing	Week 4-7	Select test system (JPetStore), design test scenarios, write JMeter scripts, execute testing and apply AI	About to start
Phase 3 - Reporting & Presentation	Week 8-9	Prepare slides, write detailed report, record demo video and conduct presentation	Not started yet

Currently the team is at the end of Phase 1, preparing to move to Phase 2 with test system selection and business flow analysis.

Chapter 2

Phase 1: Foundation & Research

2.1 Planning

2.1.1 Phase 1: Foundation & Research (Week 1-3)

Task	Content	Output	Week	Assignee
Lập kế hoạch Seminar	Xác định mục tiêu, chủ đề, phân vai nhóm	File kế hoạch (PDF/Docx)	Week 1	Cao Nhi
Tìm hiểu cơ bản về Performance Testing	Load, Stress, Spike, Endurance Testing	Slide hoặc notes tổng hợp	Week 2	Thanh Thuý
Nghiên cứu ứng dụng AI trong Performance Testing	Ví dụ: AI để phát hiện bất thường, điều chỉnh tải	Mô tả case sử dụng AI (1 trang)	Week 2	Cao Nhi
So sánh công cụ Performance Testing	JMeter, NeoLoad, WebLOAD, LoadUI, LoadRunner	Bảng so sánh (.doc/.md)	Week 2	Minh Trí
Nghiên cứu sâu JMeter (lý thuyết)	Kiến trúc, thành phần: Thread Group, Sampler, Listener...	Tài liệu hướng dẫn nội bộ	Week 3	Cát Tường
Cài đặt & cấu hình JMeter	Setup JMeter, test thử sample plan	Ảnh/clip minh họa setup + script thử nghiệm	Week 3	Việt Tú

2.1.2 Phase 2: Scenario Design & Testing (Week 4-7)

Task	Content	Output	Week	Assignee
Chọn hệ thống để test	JPetStore	Tài liệu mô tả hệ thống (1 trang)	Week 4	Cao Nhi
Phân tích luồng nghiệp vụ cần test	Đăng ký → Đăng nhập → Mua hàng → Thanh toán	Sơ đồ luồng (draw.io/pdf)	Week 4	Thanh Thuý

Task	Content	Output	Week	Assignee
Thiết kế test plan, test case (sampler)	Viết test case dạng bảng; xác định mục tiêu test (CCU, thời gian)	Test Plan (file mô tả .doc)	Week 5	Cát Tường
Viết test script bằng JMeter	Tạo script với HTTP Request, Assertion, Listener, Timer...	.jmx file	Week 5	Minh Trí
Chạy test và ghi nhận kết quả	Ghi lại các chỉ số: response time, throughput, error rate...	Bảng kết quả (.csv), biểu đồ thống kê	Week 6	Việt Tú
Nâng cao: Áp dụng AI	Tự động phân tích log, cảnh báo bất thường	Code/script, prompt	Week 7	Cao Nhi

2.1.3 Phase 3: Reporting & Presentation (Week 8-9)

Task	Content	Output	Week	Assignee
Chuẩn bị Slide trình bày	Nội dung: lý thuyết → công cụ → demo → kết luận	File .pptx	Week 8	All
Báo cáo phần lý thuyết	Trình bày khái niệm, công cụ, lý do chọn	Mục I trong file Report	Week 8	Thanh Thuý
Báo cáo phần thực hành (Kịch bản)	Mô tả kịch bản test	Mục II, phần 1 trong file Report	Week 8	Cát Tường
Báo cáo phần thực hành (Testing)	Mô tả quy trình test, kết quả	Mục II, phần 2 trong file Report	Week 8	Minh Trí
Phân tích & Đánh giá	Phân tích hiệu quả, hạn chế, đề xuất	Mục III trong file Report	Week 8	Cao Nhi
Quay video demo thực tế	Chạy test + mở báo cáo kết quả	Video .mp4 (3-5 phút)	Week 8	Việt Tú
Thuyết trình	Lý Thuyết, Kịch bản, Quy trình, Video demo		Week 9	MC (Cao Nhi) + All